Phylogenetic Analysis of Electric Organ Associated Genes in Electric Fishes

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Abstract

Electric organs have evolved independently in multiple fish lineages, making electric fishes a useful model for studying convergent evolution. Previous transcriptomic work identified electric organ (EO) associated gene clusters in Electrophorus electricus, electric eel, including genes upregulated in electric organ tissue and genes downregulated relative to skeletal muscle. In this project, I used these published EO associated genes as a biological starting point, but instead of reproducing the original transcriptome analysis, I performed an updated comparative coding DNA sequence (CDS) analysis using the latest NCBI genome annotations. I selected shared coding sequences from five species: two gymnotiform electric fishes, two mormyrid electric fishes, and Ictalurus punctatus as a non-electric outgroup. For each selected marker gene, I retained one representative CDS per species, generated codon-aware alignments using MACSE, and inferred phylogenies using distance-based, parsimony, maximum-likelihood, Bayesian, and coalescent-based approaches. Across methods, the inferred trees consistently grouped *Electrophorus electricus* (electric eel) with *Brachyhypopomus gauderio* and *Brienomyrus brachyistius* with *Paramormyrops kingsleyae*, with *Ictalurus punctatus* as the outgroup. These results suggest that the selected EO-associated coding sequences recover the expected major lineage relationships, although the small taxon set limits conclusions about convergence.

Introduction

Out of approximately 30,000 known fish species in the world, only about 350 fishes have specialized electric organs (EO), making electric fishes a relatively small but biologically remarkable group. They are mainly found in marine environments and freshwater rivers of Africa and South America. Electric fishes are generally classified as strongly electric fishes, which can generate powerful discharges used for predation or defense, or weakly electric fishes, which produce lower-voltage electric fields used mainly for communication and navigation[1].

Electric fishes provide a well-known example of convergent evolution, in which EOs have evolved independently across fish lineages. These organs generate electric fields used in navigation, communication, predation, and defense. Because many EOs are derived from skeletal muscle or muscle precursor cells, they provide a powerful system for studying how similar complex traits can arise through repeated modification of ancestral muscle programs[2].

Previous work by Gallant et al. showed that electric organs in distantly related fishes share common molecular features despite their independent origins. That study assembled the electric eel genome and compared transcriptomes from skeletal muscle and the three electric organs of *Electrophorus electricus* (electric eel): the Main EO, Hunter's, and Sachs' organs. Using k-means clustering of gene expression patterns, the authors identified EO-associated gene clusters. In particular, they highlighted cluster 9 genes, which were upregulated in electric organ tissue, and cluster 1 genes, which were downregulated in electric organs relative to skeletal muscle. These gene sets provide a biologically meaningful starting point for comparing EO-associated coding sequences across electric and non-electric fish species[2].

Electric organ function is also closely tied to ion transport, membrane excitability, and muscle-derived cellular physiology. For example, proteomic work on *E.electricus* showed that its three electric organs support different biological functions: weak discharges for navigation and communication, and stronger intermittent discharges for predation and defense. That study found differences in protein and phosphoprotein abundance among electric organs, including proteins involved in sodium and potassium transport, supporting the idea that electric organ specialization is reflected at the molecular level[3].

More recent comparative genomic work has extended this question from expression differences to molecular evolution. Wang and Yang[4] analyzed thousands of orthologs across electric and non-electric fishes and reported evidence for accelerated protein evolution, positive selection, and convergent amino acid substitutions in electric fish lineages. They highlighted genes related to membrane structure, ion channels, transmembrane transport, and electrical or muscular activity, which are all relevant to electric organ function. This suggests that electric fishes may show repeated molecular changes associated with electrogenesis.

In this project, I use the electric-organ-associated cluster 1 and cluster 9 genes from Gallant et al. as the biological starting point, but I do not use the original transcriptome analysis. Instead, I perform an updated coding-sequence-based comparative analysis using modern genome annotations. I focus on five species: *Electrophorus electricus* and *Brachyhypopomus gauderio* as gymnotiform electric fishes, *Brienomyrus brachyistius* and *Paramormyrops kingsleyae* as mormyrid electric fishes, and *Ictalurus punctatus* as a non-electric outgroup. The goal is to ask whether selected EO-associated coding sequences recover recognizable phylogenetic relationships among these taxa and whether this CDS-based view can provide a starting point for interpreting sequence conservation and divergence in genes associated with electric organ biology.

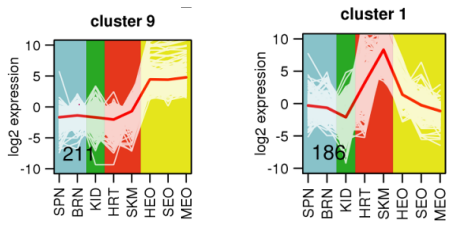

Fig 1: Clusters 9 and 1 from Gallant et al. (2014) depicting the up-regulation and down-regulation of EO genes in the EOs with respect to skeletal muscle in electric eel.

Materials and Methods

Data source and marker gene selection

This project used EO-associated genes identified by Gallant et al. as the biological starting point. I focused on cluster 9 genes, which were upregulated in electric organ tissue, and cluster 1 genes, which were downregulated in EO relative to skeletal muscle. I did not reproduce the original transcriptome analysis because the available supplementary annotations were incomplete or inconsistent for some species, and some genomes used in the original study are now obsolete. Instead, I used the published EO-associated gene clusters to define a candidate marker set for an updated comparative CDS-based phylogenetic analysis using current NCBI genome annotations.

Five species were included: *Electrophorus electricus* (`Ee`) and *Brachyhypopomus gauderio* (`Bg`) as gymnotiform electric fishes, *Brienomyrus brachyistius* (`Bb`) and *Paramormyrops kingsleyae* (`Pk`) as mormyrid electric fishes, and *Ictalurus punctatus* (`Ip`) as a non-electric outgroup. This sampling included electric fishes from two independent electric-fish lineages and one non-electric species for rooting.

For each species, I downloaded annotated coding DNA sequence FASTA files and corresponding genome annotation files from NCBI Datasets. I selected recent assemblies with RefSeq annotations when available. Gene names were parsed from CDS FASTA headers and compared case-insensitively across species. I first selected up to 100 cluster 9 genes and up to 100 cluster 1 genes, removed entries with missing or inconsistent gene names, and ranked the remaining genes using an electric-organ versus skeletal-muscle expression contrast. For cluster 9, genes were ranked by high EO-to-muscle expression; for cluster 1, genes were ranked by high muscle-to-EO expression. Duplicate gene names were removed by keeping the best-scoring entry.

This produced 182 candidate EO-associated markers: 82 from cluster 9 and 100 from cluster 1. I then identified the subset of these genes present in all five species, giving 72 shared EO-associated genes. From these, I selected the top 25 upregulated and top 25 downregulated genes, giving a final 50-gene dataset. When multiple CDS records existed for the same gene in a species, I retained the longest CDS as the representative sequence. The final input for alignment consisted of 50 per-gene FASTA files, each containing one CDS sequence from each of the five species.

Multiple Sequence Alignment (MSA)

Because the dataset consisted of protein-coding DNA sequences, I compared two multiple sequence alignment tools: Multiple Alignment using Fast Fourier Transform (MAFFT) and Multiple Alignment of Coding Sequences (MACSE). Multiple sequence alignment places homologous DNA positions across species into the same columns, allowing each site to be compared in downstream phylogenetic analyses.

MAFFT is an efficient general-purpose multiple sequence alignment program, especially with small datasets. I ran MAFFT v7.526 using the L-INS-i strategy with `--localpair --maxiterate 1000`, since the dataset had only five sequences per gene and accuracy was more important than speed[5]. However, MAFFT is not codon-aware, and some CDS alignments showed non-triplet gap patterns on inspection.

Therefore, I used MACSE v2.07 as the primary aligner. MACSE is designed for coding sequences and aligns nucleotide CDS while considering amino acid translations, frameshifts, and stop codons[6]. The main strength of MACSE is that it preserves codon structure, which is important for protein-coding genes. Its limitation is that the output can contain special frameshift or stop-codon characters that require cleanup before downstream phylogenetic analysis. MACSE was run using `alignSequences`, followed by `exportAlignment` to clean the output. Selected alignments were inspected in Jalview, a graphical alignment viewer and analysis workbench[7]. The cleaned MACSE nucleotide alignments were then reordered to keep species order consistent across genes and used for phylogenetic inference.

Distance-based and Parsimony-based Phylogenetic Inference

I used distance-based and parsimony-based methods as initial tree-building approaches on the cleaned MACSE nucleotide alignments. These analyses were done in R using `ape`[8] and `phangorn`[9]. Distance methods convert aligned sequence differences into pairwise distances before inferring a tree, while parsimony methods search for the tree requiring the fewest nucleotide changes.

For distance-based inference, I calculated pairwise DNA distances using `dist.dna()` and inferred BioNJ trees using `bionj()` from `ape`. BioNJ was chosen because it does not assume a strict molecular clock. I tested F84 and TN93 models and used TN93 as the main model because it allows unequal base frequencies and different transition rates. The strength of distance methods is that they are simple and computationally efficient, making them useful for exploratory tree inference. Their limitation is that they compress each alignment into pairwise distances, which loses site-level information.

For parsimony inference, I used `optim.parsimony()` from `phangorn` with Fitch parsimony and SPR rearrangements. Fitch parsimony treats all nucleotide substitutions as equal cost, and SPR searches tree space more broadly while remaining feasible for five taxa. I also used `bab()`, an exact branch-and-bound search, to check the parsimony scores. The strength of

parsimony is that it is intuitive and does not require an explicit substitution model. Its limitation is that it can be sensitive to homoplasy, unequal evolutionary rates, and long-branch attraction.

For both approaches, I inferred one tree per gene and summarized the gene trees using majority-rule consensus trees with a threshold of 0.8. This means that a clade was included in the consensus tree if it appeared in at least 80% of the gene trees. The trees were rooted with *I.punctatus* as the outgroup and ladderized only for visualization. These methods were useful as simple comparisons to the model-based approaches.

Maximum-likelihood Phylogenetic Inference

I used IQ-TREE[10] for maximum-likelihood phylogenetic inference. Maximum likelihood is a model-based approach that estimates the tree topology, branch lengths, and substitution model parameters that make the observed sequence alignment most probable. IQ-TREE was chosen because it supports partitioned multi-gene analyses, automatic model selection, and fast branch-support estimation. However, its results do depend on the model fit and heuristic tree search.

I ran IQ-TREE on both the cleaned nucleotide and amino acid MACSE alignments using a gene-partitioned analysis, where each gene was treated as a separate partition. I used `MFP+MERGE`, which performs ModelFinder Plus model selection and then merges partitions with similar evolutionary patterns to reduce over-parameterization. Branch support was estimated using 1000 ultrafast bootstrap replicates and 1000 SH-aLRT replicates. Ultrafast bootstrap gives a fast approximation of standard bootstrap support, while SH-aLRT tests whether each branch is significantly better supported than the nearest alternative branch. I used `Ip` as the outgroup for visualization, although IQ-TREE reports maximum-likelihood trees as unrooted.

Bayesian Phylogenetic Inference

I used MrBayes for Bayesian phylogenetic inference. Bayesian phylogenetics uses Markov chain Monte Carlo (MCMC) sampling to estimate a posterior distribution of trees and model parameters, and clade support is reported as posterior probability rather than bootstrap support. A strength of MrBayes is that it estimates uncertainty directly through posterior probabilities and can model different evolutionary patterns across partitions. Its limitation is that results depend on prior/model choices and adequate MCMC convergence[11].

The cleaned MACSE nucleotide alignments were concatenated into a MrBayes-compatible NEXUS file using FASconCAT-G[12]. I used codon-position partitioning instead of gene-level partitioning because the data were coding sequences and the dataset had only five taxa. The alignment was divided into first, second, and third codon positions. I used `nst=mixed` and `rates=gamma`, allowed base frequencies and gamma shape parameters to vary among codon positions, and allowed partition rates to vary.

The full MrBayes run used two independent runs, four chains per run, and 1,000,000 generations, sampling every 1,000 generations. I discarded the first 25% of samples as burn-in.

Convergence was assessed using split-frequency diagnostics, PSRF values, and trace inspection. The final consensus tree was summarized using sumt, with posterior probabilities used as node-support values.

Coalescent-based Species-tree Inference

I used the Accurate Species TRee ALgorithm (ASTRAL-IV) from the ASTER package for coalescent-based species-tree inference[13]. Unlike concatenated IQ-TREE and MrBayes analyses, ASTRAL estimates a species tree from a set of gene trees under the multi-species coalescent framework. A strength of ASTRAL-IV is that it can account for gene-tree discordance caused by incomplete lineage sorting. Its limitation is that it depends on the accuracy of the input gene trees and does not directly analyze the original sequence alignments.

For this analysis, I first inferred one maximum-likelihood tree per gene using IQ-TREE and the cleaned MACSE nucleotide alignments. The resulting 50 gene trees were combined into a single input file for ASTRAL-IV. I used Ip as the outgroup for rooting and used the more exhaustive search settings `-r 16 -s 16`, which was recommended for datasets with a small number of taxa and limited gene set. ASTRAL branch support values were interpreted as local posterior probabilities.

Tree Visualization and Comparison

Final trees were visualized in R using ggtree, which allows phylogenetic trees to be plotted and annotated with additional metadata. Trees were rooted with Ictalurus punctatus as the outgroup for biological interpretation and ladderized only to improve readability. Tips were colored by major lineage: gymnotiform electric fishes, mormyrid electric fishes, and the non-electric outgroup.

I compared the final IQ-TREE and MrBayes topologies using the Robinson–Foulds (RF) distance and cophylogeny-style plots. The RF distance measures topological differences between two trees by comparing their splits, while the cophylogeny plot places two trees side by side and connects matching taxa. These visual comparisons were used to check whether different inference methods recovered the same major relationships.

Results

The final dataset consisted of 50 EO-associated marker genes shared across all five species. These markers were selected from 72 shared candidate genes identified from the original cluster 1 and cluster 9 EO-associated gene sets. For each gene, one representative CDS sequence was retained per species, producing 50 per-gene FASTA files and 250 sequences overall.

Because these were coding DNA sequences, I used MACSE-cleaned nucleotide alignments as the main input for phylogenetic inference. The final concatenated nucleotide dataset used for model-based analyses contained 5 taxa and 107,499 sites, while the translated amino acid dataset contained 5 taxa and 35,833 sites. In the final IQ-TREE input, both datasets were reported as having 0% missing data.

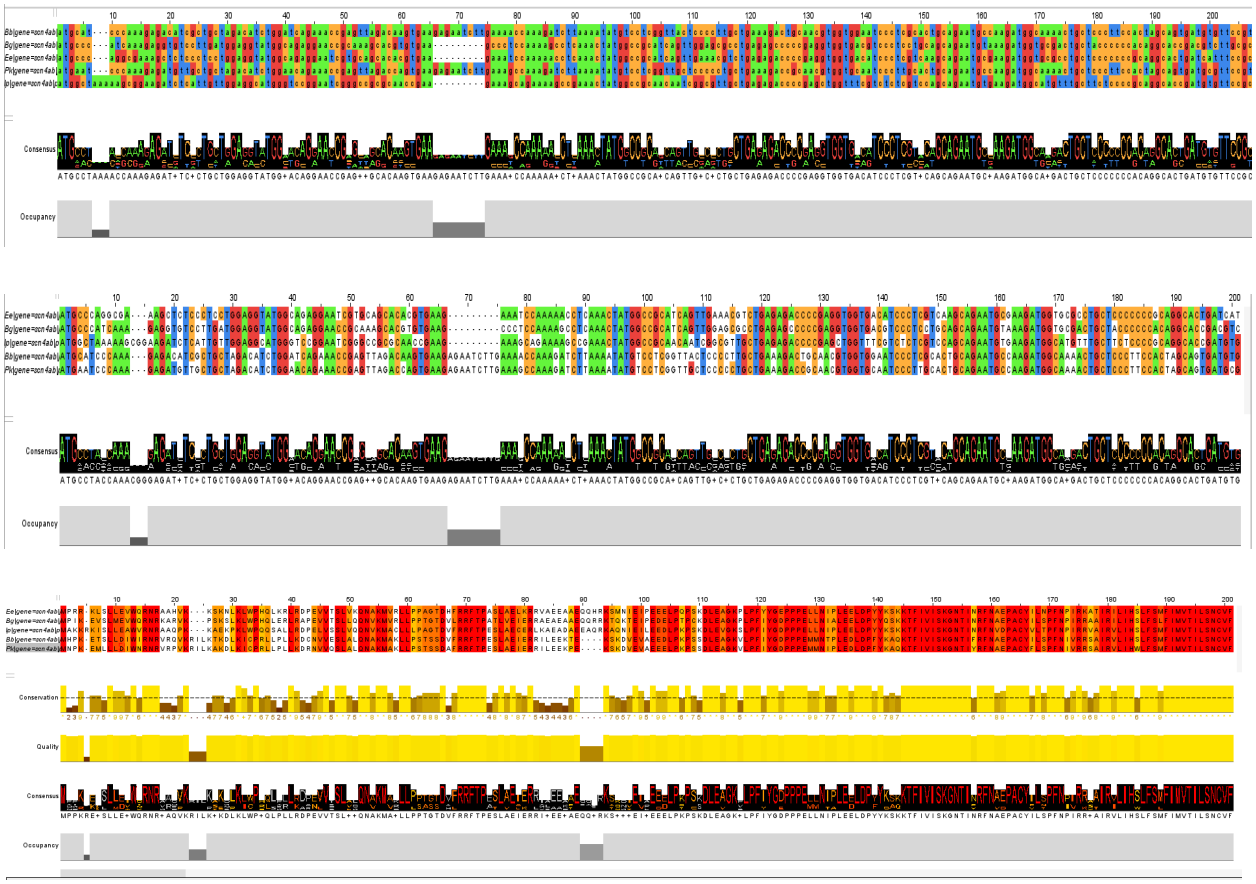

Fig 2: The first 200bp of aligned sequences for an up-regulated gene scn4ab as outputted from a) MAFFT, b) MACSE nucleotide sequence, and c) MACSE translated amino acid sequences across all 5 species and viewed in JALView

A major result was that all inference methods recovered the same broad species relationships. This consistency suggests that, for a dataset of this size and taxon sampling, the choice of tree-inference method had little effect on the final topology.

Distance-based and parsimony-based results

The distance-based BioNJ analyses using both F84 and TN93 distance models recovered the same overall consensus topology. In both cases, *Electrophorus electricus* grouped with *Brachyhypopomus gauderio*, and *Brienomyrus brachyistius* grouped with *Paramormyrops kingsleyae*, with *Ictalurus punctatus* placed as the outgroup after rooting. The agreement

between F84 and TN93 suggests that the distance-based consensus result was not strongly sensitive to the selected distance model.

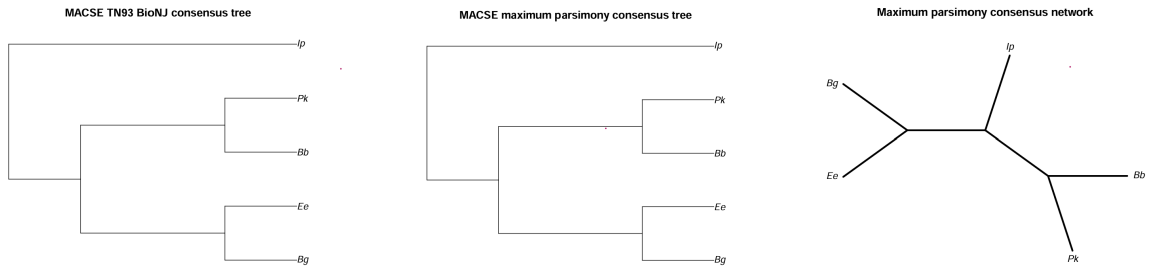

Fig 3: The trees consensus trees inferred through a) distance and b) parsimony methods; and c) the consensus network inferred through parsimony

The maximum-parsimony analysis recovered the same major relationships. The per-gene parsimony trees were summarized using a consensus tree, and the branch-and-bound search produced the same parsimony scores as the optimized trees, indicating that the optimized parsimony searches had reached the best score for these alignments. The parsimony consensus network was also largely tree-like and did not show strong reticulation, suggesting little major conflict among the per-gene parsimony trees at the chosen split-frequency threshold.

Overall, both simpler exploratory approaches recovered the same biologically expected grouping: the two gymnotiform electric fishes grouped together, the two mormyrid electric fishes grouped together, and the non-electric channel catfish was separated as the outgroup.

Maximum-likelihood results

The IQ-TREE maximum-likelihood analyses were run separately on nucleotide and amino acid alignments. Both analyses recovered the same topology:

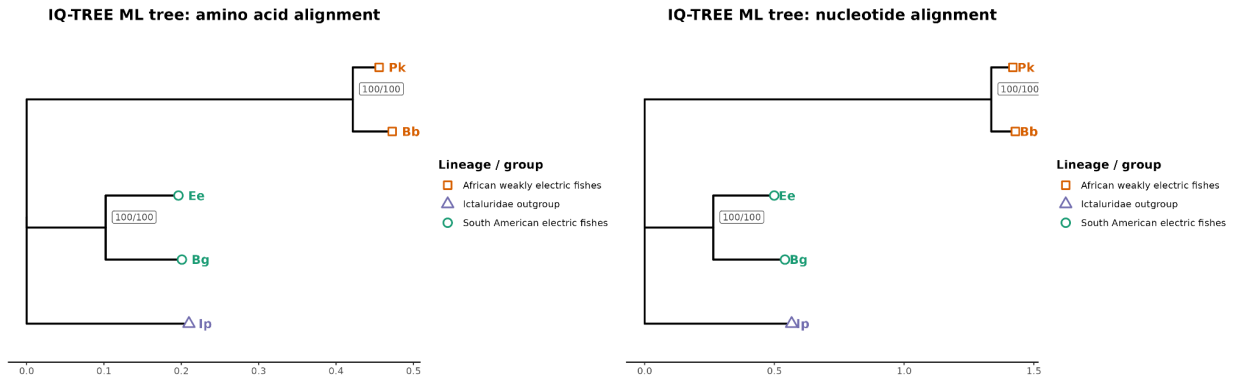

Fig 4: The trees consensus trees inferred through IQ-TREE using a) amino acid and b) nucleotide MACSE alignments

In the nucleotide analysis, IQ-TREE reported 5 taxa, 17 partitions, and 107,499 sites. In the amino acid analysis, it reported 5 taxa, 13 partitions, and 35,833 sites. Both trees recovered *Bg* + *Ee* and *Bb* + *Pk* as the two major ingroup clades. Each internal clade had 100/100 support, where the first value is SH-aLRT support and the second value is ultrafast bootstrap support.

The nucleotide and amino acid trees had the same topology but different branch lengths. This is expected because nucleotide sequences contain more sites and include synonymous substitutions, while amino acid sequences represent protein-level changes and are usually more conservative. Because the nucleotide analysis included one partition with unusually long branches, I treated the amino acid tree as the primary ML result wherever applicable.

Bayesian phylogenetic results

The MrBayes analysis also recovered the same major topology as IQ-TREE. The Bayesian consensus tree supported a close relationship between *Bg* and *Ee* and between *Bb* and *Pk*, with *Ip* as the outgroup. The major clades had posterior probabilities of 1.00, indicating very strong posterior support.

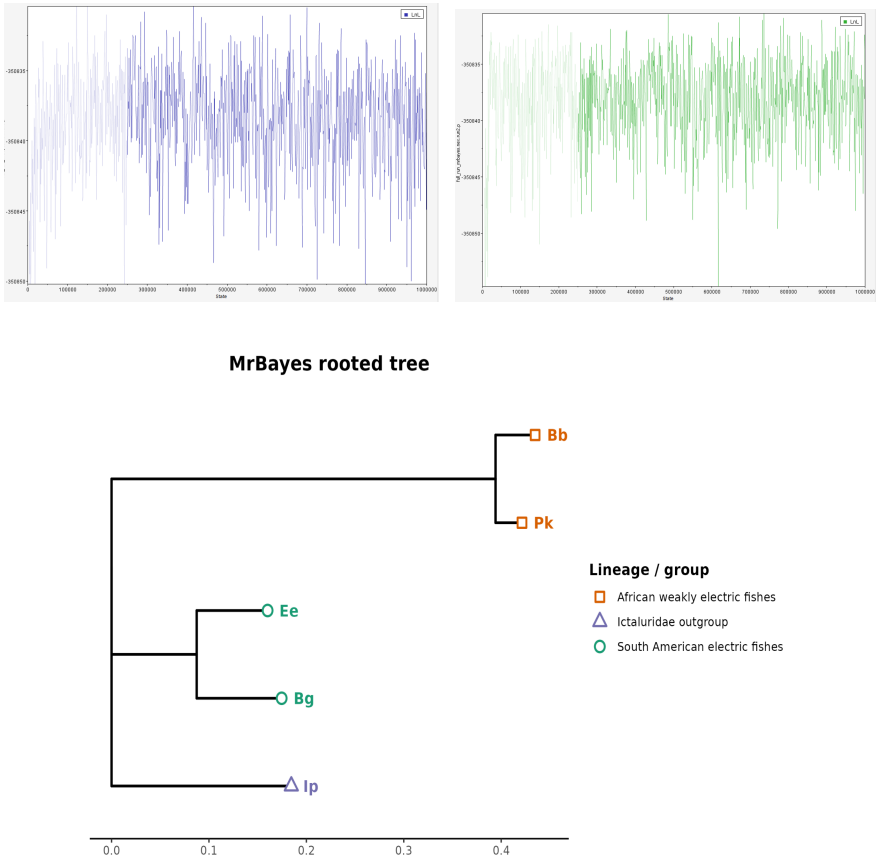

Fig 5: The top 2 plots, a) and b) show the log-likelihood trace plots for the runs 1 and 2. The plot c) in the bottom shows the rooted tree inferred from MrBayes

The MCMC run showed good convergence. The analysis used two independent runs with four chains each for 1,000,000 generations, with a 25% burn-in. The average standard deviation of split frequencies was 0.000000, and PSRF values were close to 1.0, indicating that the chains mixed well in each run and that the independent runs sampled similar posterior distributions. Most ESS values were above 400, and the few lower values were still above 100, so the run was considered sufficient for this analysis.

The MrBayes parameter estimates also matched expectations for coding sequences. Codon position 3 evolved fastest, codon position 1 was intermediate, and codon position 2 evolved slowest, consistent with stronger functional constraints on first and second codon positions.

Coalescent-based species-tree results

The ASTRAL-IV species-tree analysis used 50 per-gene IQ-TREE trees as input. ASTRAL-IV recovered the same major topology as the concatenated IQ-TREE and MrBayes analyses.

This topology placed *Ip* as the outgroup and separated the electric fishes into two major lineages: *Ee* + *Bg* as gymnotiform electric fishes and *Bb* + *Pk* as mormyrid electric fishes. Both major ingroup clades had local posterior probability support of 1.0.

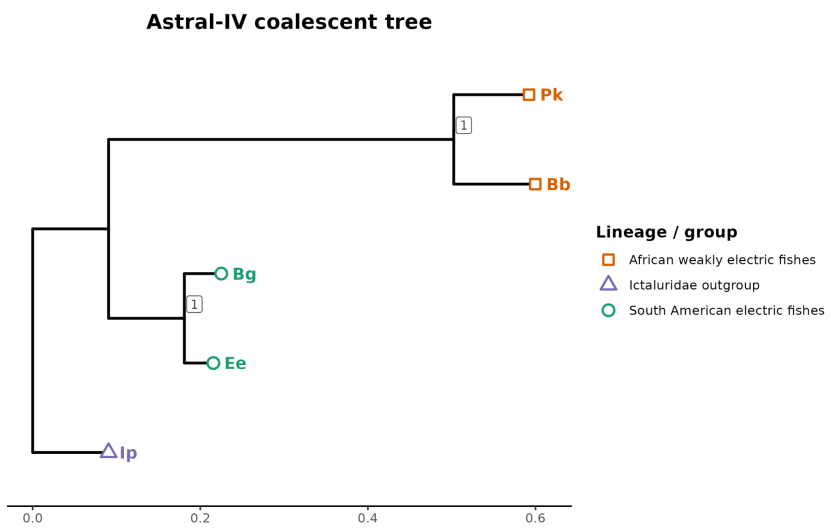

Fig 6:This figure shows inferred species tree through the coalescent method ATRAL-IV

The ASTRAL-IV tree had the same topology as the concatenated analyses, but branch lengths differed across methods. This is expected because IQ-TREE and MrBayes estimate branch lengths from concatenated sequence alignments, whereas ASTRAL-IV estimates a species tree from independently inferred gene-tree topologies under a coalescent framework.

Comparison across methods and biological interpretation

Across all the tree inference analyses mentioned above, the same major relationship was recovered: *Electrophorus electricus* grouped with *Brachyhypopomus gauderio*, *Brienomyrus brachyistius* grouped with *Paramormyrops kingsleyae*, and *Ictalurus punctatus* was placed as the outgroup. The MrBayes and ASTRAL-IV topologies were identical, with a Robinson–Foulds distance of 0, meaning the two trees had the same topology. The recovery of the same major species relationships across distance, parsimony, maximum-likelihood, Bayesian, and coalescent-based approaches suggests that, for this dataset, the choice of inference method had little effect on the final topology. This may be partly because the dataset was small and highly constrained, with only 5 taxa, 50 genes, and 250 total sequences.

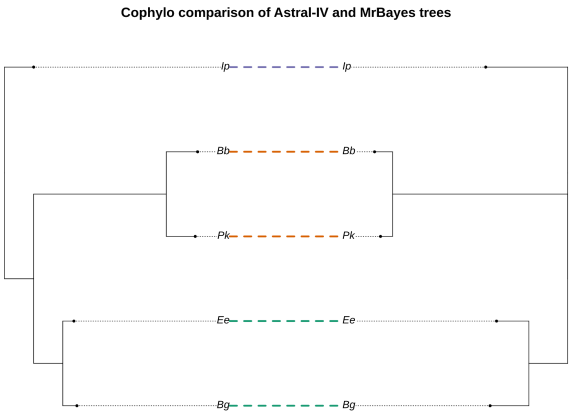

Fig 7:This figure shows the cophylogeny between ASTRAL-IV and MrBayes inferred species trees

Biologically, the recovered topology is consistent with the known relationships among the sampled fishes. *E. electricus* and *B. gauderio* are gymnotiform electric fishes from South America, whereas B. *brachyistius* and *P. kingsleyae* are mormyrid electric fishes from Africa. Therefore, the selected EO-associated coding sequences recovered broad lineage relationships rather than grouping all electric fishes together based on the convergent electric-organ phenotype. This suggests that, in this marker set, the dominant phylogenetic signal reflects shared ancestry among major lineages, even though the genes were selected because of their association with electric organ biology.

Discussion

This project used EO-associated genes from published transcriptomic work as a starting point for an updated CDS-based phylogenetic analysis. Across distance, parsimony, maximum-likelihood, Bayesian, and coalescent-based methods, the same major topology was recovered. This agrees with the expected separation of South American gymnotiform electric

fishes and African mormyrid electric fishes, suggesting that these EO-associated coding sequences retain a strong lineage-level phylogenetic signal.

The agreement across methods increases confidence that the main relationships recovered in this analysis are stable for the selected dataset. However, it also reflects an important limitation of the chosen data. Because the dataset included only five taxa, 50 genes, and 250 total sequences, there may not have been enough taxonomic or sequence complexity for the different inference methods to produce strongly different topologies. In this context, the consistent topology suggests that method choice had little effect on the final tree, rather than proving that all methods would behave similarly in a larger phylogenomic dataset.

The small taxon set also limits the biological conclusions that can be drawn about convergent evolution. The analysis included two gymnotiform electric fishes, two mormyrid electric fishes, and one non-electric outgroup, which was sufficient to recover the expected broad lineage relationships. However, it is not enough to test whether EO-associated genes show repeated molecular changes across many independent electrogenic lineages. A more informative analysis would include additional taxa from multiple electric and non-electric lineages, more outgroups, and a larger gene set.

Another limitation is that this project compared genome-derived CDS annotations rather than directly analyzing transcriptome sequences from electric organ and skeletal muscle tissues. Using current genome annotations made the analysis more standardized across species, but it also moved the project away from the expression-based context in which the original EO-associated clusters were identified. Future work could improve this by incorporating transcriptome-derived sequences or expression-supported isoforms from electric organ tissues, which would better connect the phylogenetic analysis to electric organ biology.

Finally, the marker set was restricted to selected EO-associated genes, and gene-name matching across genome annotations may miss true orthologs or include imperfect matches. Choosing the longest CDS per gene also simplified transcript diversity and may not always represent the most biologically relevant isoform. Overall, this analysis shows that selected EO-associated CDS markers recover the expected major relationships among the sampled taxa, but stronger conclusions about molecular convergence would require broader taxon sampling, expanded gene sampling, explicit orthology inference, transcriptome-based comparisons, and direct tests for selection or convergent amino acid substitutions.

Code Availability

All scripts, markdown workflow notes, intermediate processing descriptions, and result files for this project are available in the project GitHub repository: https://github.com/svenkatesh06/Botany653

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